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EXPERIMENT 4

DETECT SESSION HIJACKING

AIM:

To detect possible session hijacking by analysing network traffic and application logs using Wireshark and other security monitoring tools.

TOOLS REQUIRED:

1. **Kali Linux:** Used as the monitoring machine for network traffic analysis.
2. **Metasploitable2:** Used as the target machine in the isolated lab network.
3. **VirtualBox:** Used to run both virtual machines.
4. **Wireshark:** Used to capture and analyse ARP and ICMP packets.
5. **Terminal utilities:** ping, arp -a, and ip neigh to verify connectivity and inspect IP-to-MAC address mappings.
6. **DVWA:** Used in the lab to examine web session security, if required.

PROCEDURE:

1. Start Kali Linux and Metasploitable2 in VirtualBox and ensure both machines are connected to the same isolated host-only network.
2. Identify the IP addresses of the machines and verify connectivity using the ping command.

   ping -c 4 192.168.56.102
3. Open Wireshark in Kali Linux and start capturing network traffic on the eth0 interface.
4. Apply the display filter arp to examine Address Resolution Protocol requests and replies.
5. Apply the display filter icmp to observe ICMP echo requests and replies between Kali Linux and Metasploitable2.
6. Inspect the ARP neighbour table using the following command:

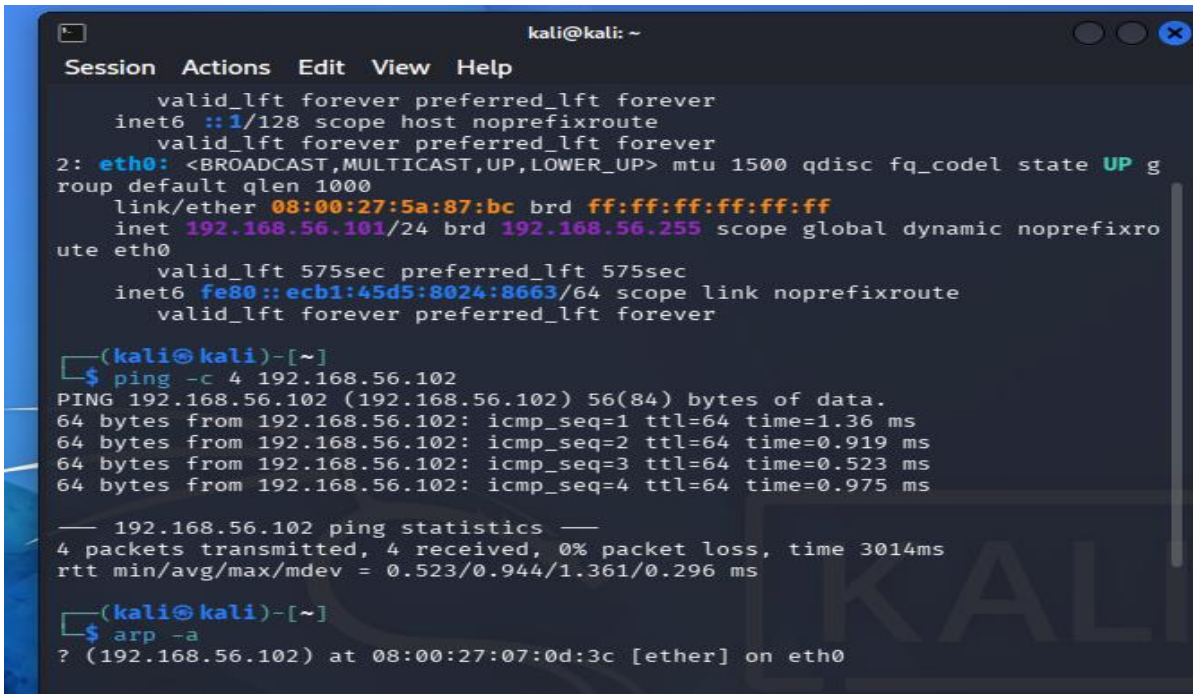
   ip neigh

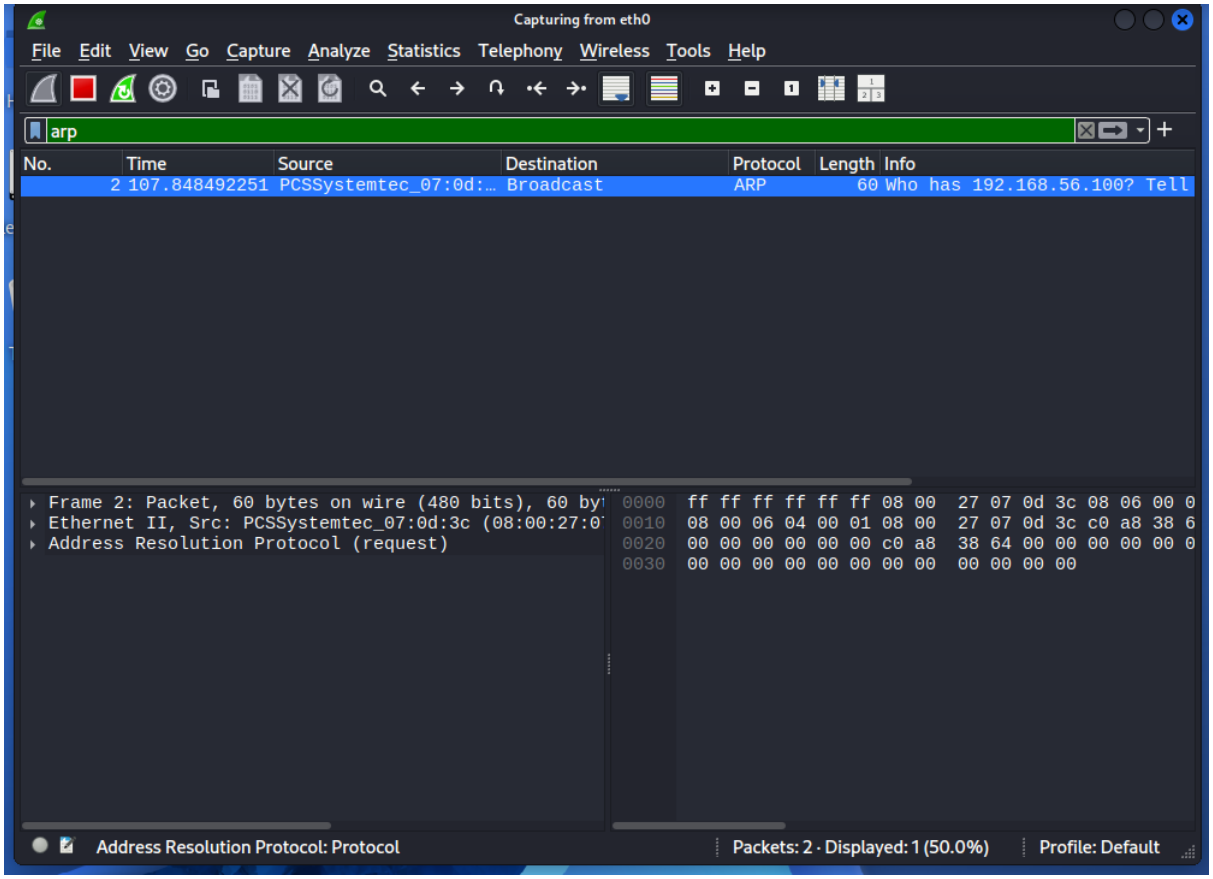
Alternatively, use arp -a to view available IP-to-MAC address mappings.

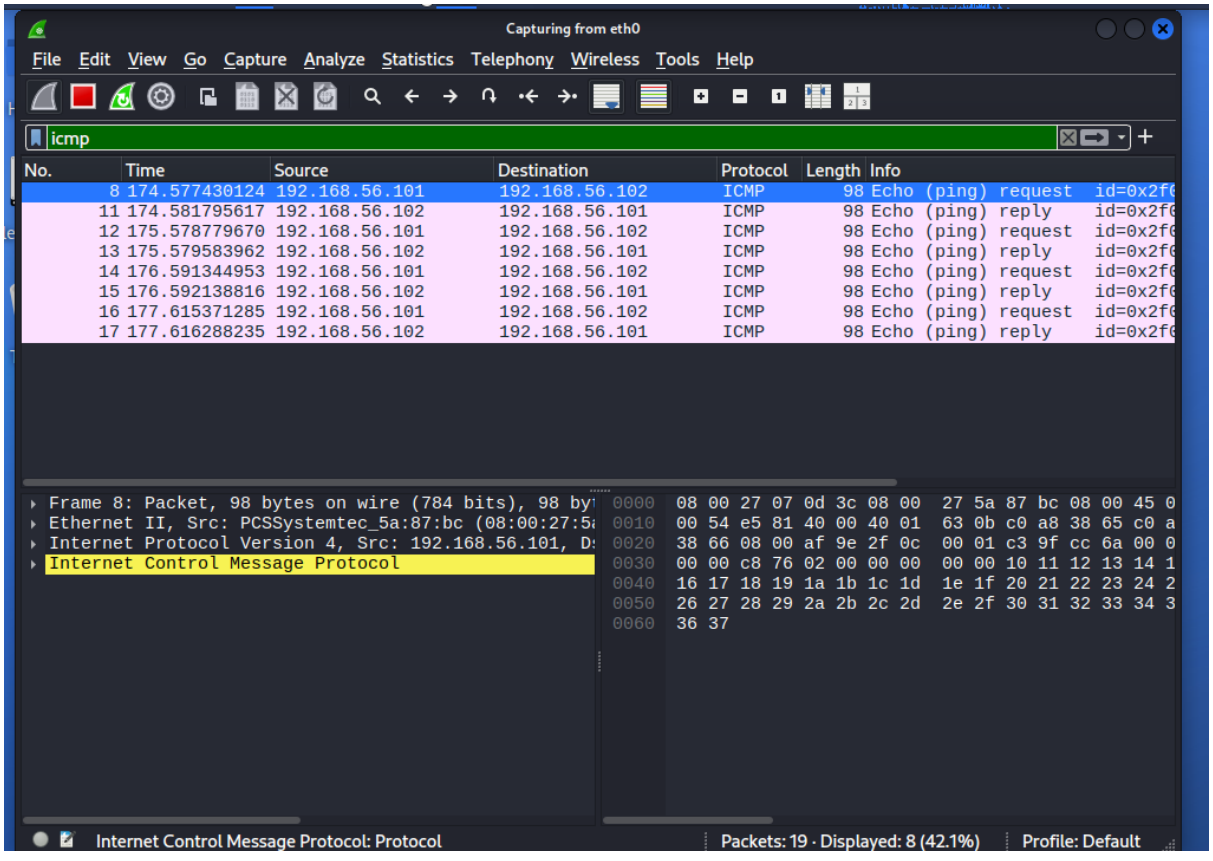
7. Compare the observed IP-to-MAC mappings and examine the captured traffic for unexpected or duplicate ARP replies that could indicate ARP spoofing.
8. Record the captured packets, connectivity results, and ARP neighbour information, and document whether any session hijacking indicators were confirmed.

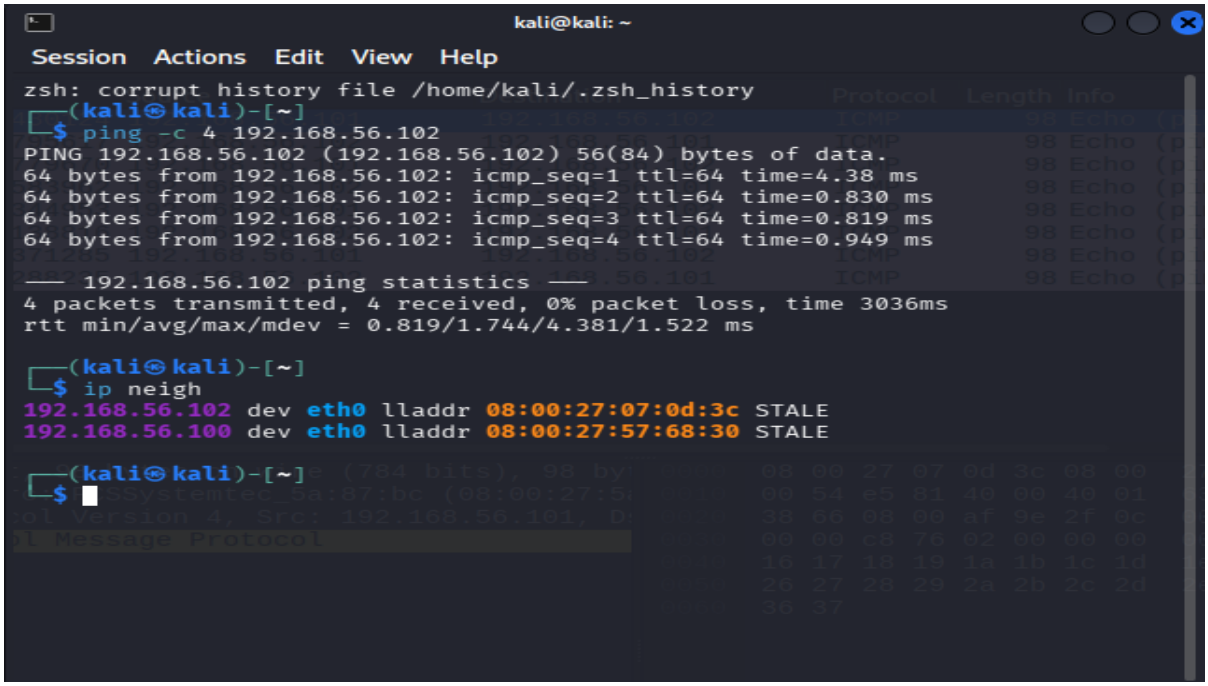
OBSERVATION:

TEST PPERFORMED	OBSERVATION
Checked IP addresses	Kali: 192.168.56.101; Metasploitable2: 192.168.56.102
Tested connectivity using ping	4 packets transmitted, 4 received, 0% packet loss.
Captured ARP packets using Wireshark	An ARP request asking for 192.168.56.100 was observed.
Analysed ICMP traffic	Echo requests and replies between Kali and Metasploitable2 were captured
Checked the ARP neighbour table using ip neigh	Metasploitable2 was mapped to MAC address 08:00:27:07:0d:3c, with state STALE.
Checked for session hijacking indicators	No confirmed ARP spoofing or session hijacking was observed in the collected evidence.




Capturing from eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
arp
No. Time Source Destination Protocol Length Info
2 107.848492251 PCSSystemtec_07:0d:… Broadcast ARP 60 Who has 192.168.56.100? Tell
Frame 2: Packet, 60 bytes on wire (480 bits), 60 by
Ethernet II, Src: PCSSystemtec_07:0d:3c (08:00:27:0
Address Resolution Protocol (request)
0000 ff ff ff ff ff ff 08 00 27 07 0d 3c 08 06 00 0
0010 08 00 06 04 00 01 08 00 27 07 0d 3c c0 a8 38 6
0020 00 00 00 00 00 00 c0 a8 38 64 00 00 00 00 00 0
0030 00 00 00 00 00 00 00 00 00 00 00 00
Address Resolution Protocol: Protocol
Packets: 2 · Displayed: 1 (50.0%)
Profile: Default


Capturing from eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Length Info
8 174.577430124 192.168.56.101 192.168.56.102 ICMP 98 Echo (ping) request id=0x2f
11 174.581795617 192.168.56.102 192.168.56.101 ICMP 98 Echo (ping) reply id=0x2f
12 175.578779670 192.168.56.101 192.168.56.102 ICMP 98 Echo (ping) request id=0x2f
13 175.579583962 192.168.56.102 192.168.56.101 ICMP 98 Echo (ping) reply id=0x2f
14 176.591344953 192.168.56.101 192.168.56.102 ICMP 98 Echo (ping) request id=0x2f
15 176.592138816 192.168.56.102 192.168.56.101 ICMP 98 Echo (ping) reply id=0x2f
16 177.615371285 192.168.56.101 192.168.56.102 ICMP 98 Echo (ping) request id=0x2f
17 177.616288235 192.168.56.102 192.168.56.101 ICMP 98 Echo (ping) reply id=0x2f
Frame 8: Packet, 98 bytes on wire (784 bits), 98 by
Ethernet II, Src: PCSSystemtec_5a:87:bc (08:00:27:5
Internet Protocol Version 4, Src: 192.168.56.101, D
Internet Control Message Protocol
0000 08 00 27 07 0d 3c 08 00 27 5a 87 bc 08 00 45 0
0010 00 54 e5 81 40 00 40 01 63 0b c0 a8 38 65 c0 a
0020 38 66 08 00 af 9e 2f 0c 00 01 c3 9f cc 6a 00 0
0030 00 00 c8 76 02 00 00 00 00 00 10 11 12 13 14 1
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 2
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 3
0060 36 37
Internet Control Message Protocol: Protocol
Packets: 19 · Displayed: 8 (42.1%)
Profile: Default



RESULT:

Network traffic was captured and analysed using Wireshark. Normal ARP requests and ICMP echo request/reply packets were observed. The ARP neighbour table was examined to understand IP-to-MAC address mapping and identify potential indicators of ARP spoofing. No session hijacking was confirmed during the observation.